Introduction

This project aims to design and implement a complete Enterprise ETL Extract, Transform, Load) solution using SQL Server Integration Services (SSIS) for a retail company. The solution processes operational data from multiple business processes including Sales, Online Sales, Inventory, and related dimensions, transforming it into a structured Data Warehouse optimized for analytical reporting and business intelligence.

The project follows enterprise ETL best practices, implementing both Full Load and Incremental Load processes while ensuring scalability, maintainability, performance, and data quality.

Project Architecture

Staging Package - Copies data from source to staging

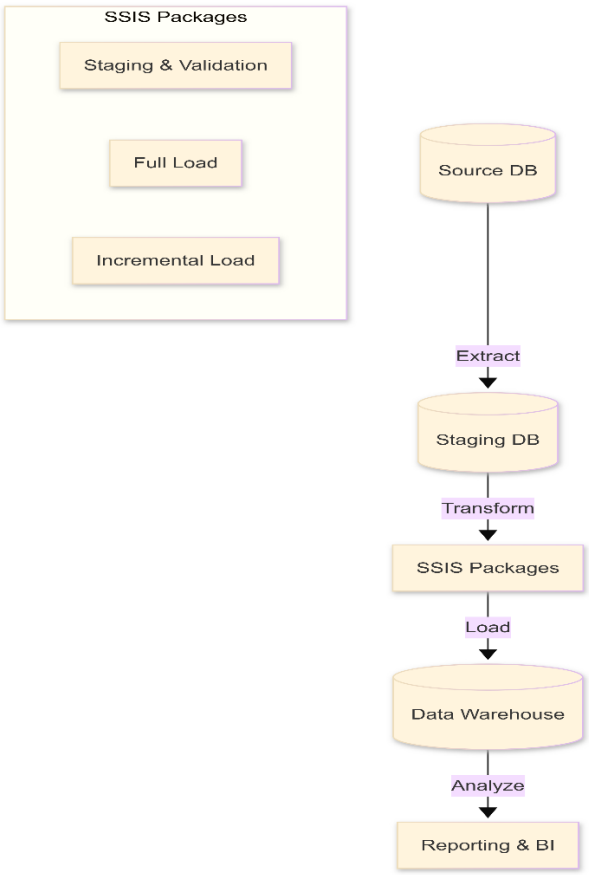
Validation Package - Validates data quality

Transformation Package - Applies business rules

Full Load Package - Initial data load

Incremental Load Package - Updates and new records

Logging Package - Tracks execution history



ETL Design

3.1 SSIS Solution Architecture

The ETL solution is divided into the following logical packages, executed in a specific sequence:

Staging Package: Extracts raw data from the source database (ContosoRetailDW) and loads it into staging tables without any transformations.

Validation Package: Applies data quality rules (Mandatory fields, numeric ranges, business rules). Invalid records are redirected to an Error Log table, while valid records proceed to the next stage.

Transformation Package: Applies business logic, financial calculations (e.g., NetSalesAmount, GrossProfit), and data standardization (e.g., TRIM, UPPER). Audit columns (LoadDate, PackageName) are added here.

Full Load Package: Truncates the target Data Warehouse tables and performs an initial bulk load of all historical data.

Incremental Load Package: Detects new and updated records using Primary Key comparisons and applies Upsert (Update/Insert) operations to the Data Warehouse.

Logging Package: Tracks the execution status, duration, and record counts of all packages into a centralized ETL_Log table.

3.2 Data Flow Design

The core data movement is handled within SSIS Data Flow Tasks using the following components:

OLE DB Source: Connects to the validated and transformed staging tables.

Derived Column: Used to create new calculated columns (e.g., ProfitMarginPercent, TotalInventoryValue) and standardize string data.

Lookup Transformation: Checks the target Data Warehouse table against the incoming data using the Primary Key to identify existing records.

Conditional Split: Routes the data into two paths based on the Lookup result:

New Records: Rows that do not exist in the destination.

Existing Records: Rows that need to be checked for updates.

OLE DB Command: Executes an UPDATE statement for modified records.

OLE DB Destination: Performs a fast, bulk INSERT for new records.

3.3 Incremental Load Strategy (Upsert Logic)

To handle incremental changes efficiently, the project implements a Slowly Changing Dimension (SCD) Type 1 approach for facts and dimensions:

For Standard Tables: The package uses the Lookup + Conditional Split + OLE DB Command/Insert pattern to separate inserts from updates.

For Large Fact Tables (e.g., FactInventory): To overcome memory pressure and optimize performance, the design utilizes the Execute SQL Task with a T-SQL MERGE statement. This performs the Upsert operation directly within the SQL Server Database Engine, bypassing SSIS memory buffers and significantly reducing execution time.

3.4 Performance Optimization & Error Handling

Buffer Tuning: The DefaultBufferMaxRows and DefaultBufferSize properties were optimized to prevent memory exhaustion during large data flows.

Fast Load: The OLE DB Destinations are configured with "Fast Load" options and appropriate batch sizes to maximize throughput.

Error Redirection: The Validation package utilizes SSIS Error Outputs to capture and log rejected rows into a dedicated ErrorLog table, ensuring that bad data never corrupts the Data Warehouse while allowing the package to continue processing valid records.

Validation Rules

Mandatory Fields Validation

- ProductKey, CustomerKey, StoreKey must not be NULL

- SalesAmount must have a value

- DateKey must be valid

Numeric Values Validation

- SalesAmount >= 0 (no negative sales)

- SalesQuantity > 0

- UnitCost >= 0

- DiscountAmount between 0 and SalesAmount

Business Rules Validation

- OrderDate cannot be in the future

- ReturnQuantity <= SalesQuantity

- Discount percentage cannot exceed 100%

- Stock quantity cannot be negative

Error Handling

Invalid records are redirected to ErrorLog table with:

- RecordID
- SourceTable
- ErrorReason
- ErrorDate
- PackageName

Transformation Rules

Financial Calculations

1. NetSalesAmount = SalesAmount - DiscountAmount

2. GrossProfit = SalesAmount - TotalCost

3. ProfitMarginPercent =
   (SalesAmount - TotalCost) /
   NULLIF(SalesAmount, 0) * 100

4. TotalInventoryValue = OnHandQuantity * UnitCost

Business Classifications

1. OrderSize =
   CASE
     WHEN SalesQuantity > 50 THEN 'Large'
     WHEN SalesQuantity > 10 THEN 'Medium'
     ELSE 'Small'
   END

2. StockStatus =

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  CASE
    WHEN OnHandQuantity < 10 THEN 'Critical'
    WHEN OnHandQuantity < 50 THEN 'Low'
    ELSE 'Sufficient'
  END
```

3. CustomerTier =

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  CASE
    WHEN TotalPurchases > 10000 THEN 'Premium'
    WHEN TotalPurchases > 1000 THEN 'Regular'
    ELSE 'New'
  END
```

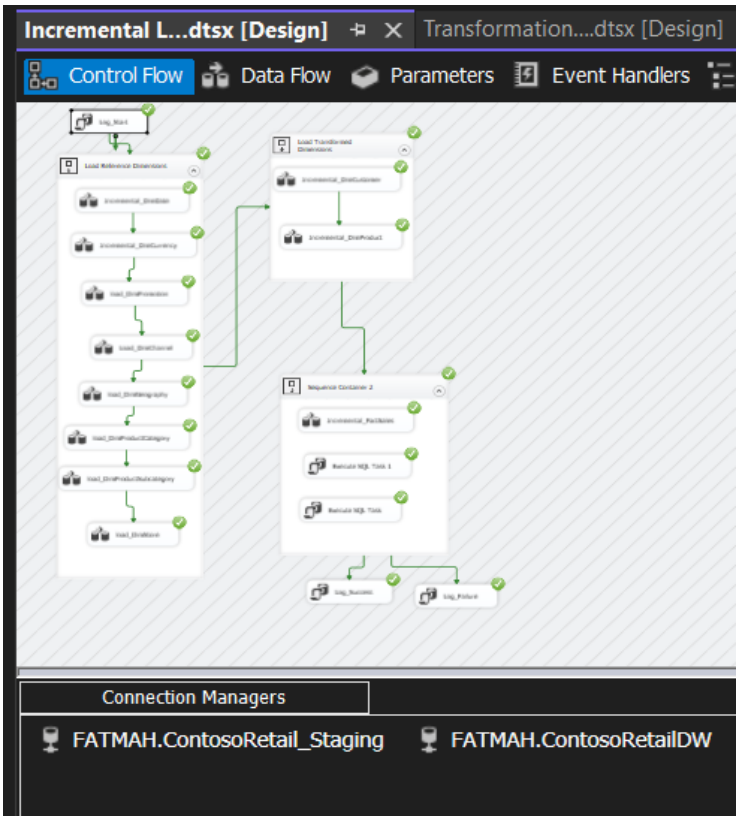
Data Standardization

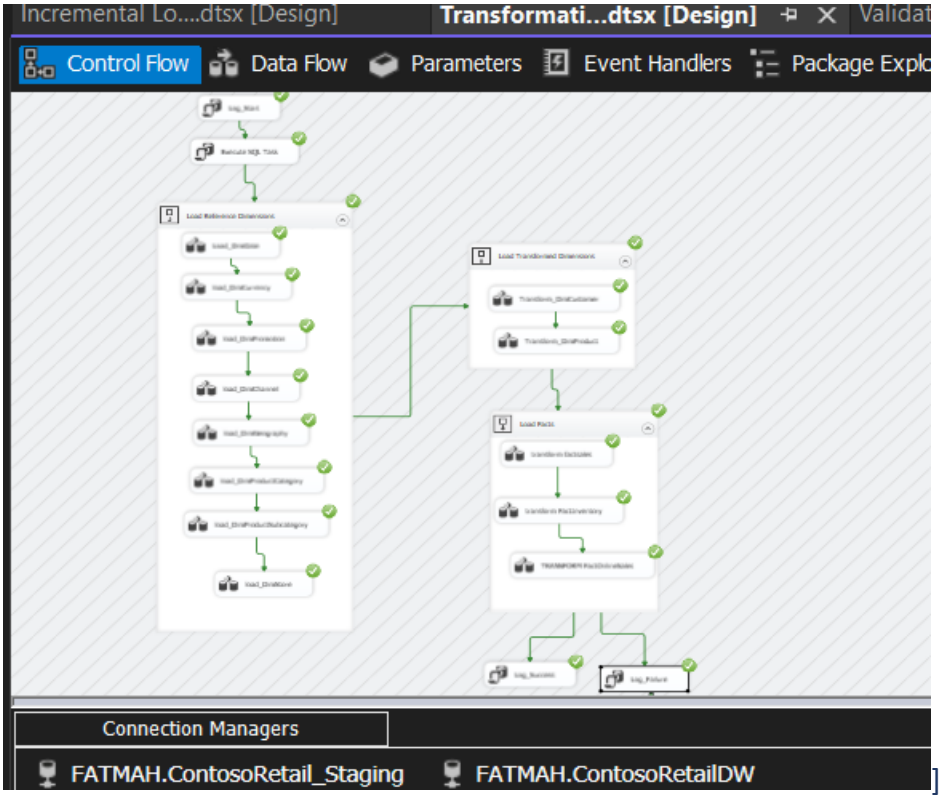
- UPPER(TRIM(ProductName)) for product names
- TRIM(CustomerName) for customer names
- ISNULL(ProductDescription, 'No Description')
- Standardized date formats

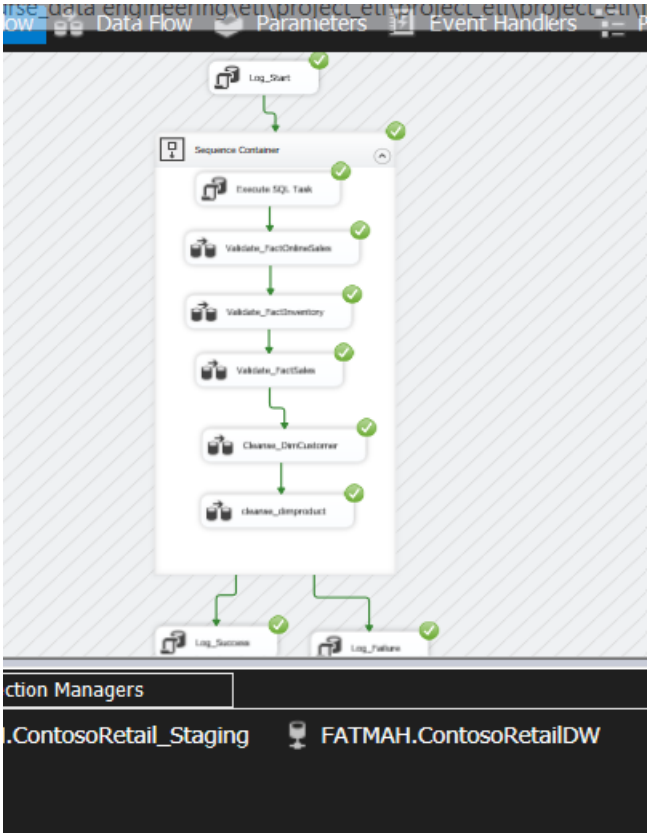
Audit Information

- LoadDate = GETDATE()
- PackageName = @[System::PackageName]
- ETLUser = @[System::UserName]
- BatchID = @[System::ExecutionInstanceGUID

Package Screenshots







Error Handling Examples (Before / After)

Before Optimization (The Problem): During the initial design of the Incremental_FactInventory package, we used the standard SSIS Data Flow approach, which relied on a Lookup Transformation to identify new and updated records.

The Issue: The FactInventory table contains a massive amount of data (over 8 million rows). The Lookup component attempted to load the entire reference table into the system's RAM using "Full Cache" mode.

The Error: This caused severe memory exhaustion on the server, leading to the following fatal error and causing the package to crash:

"The buffer manager failed a memory allocation call... The system was low on virtual memory, but was unable to swap out any buffers."

After Optimization (The Solution): To resolve this critical performance bottleneck, we completely redesigned the incremental load strategy for large Fact tables.

The Solution: We replaced the memory-heavy SSIS Lookup and Data Flow components with an Execute SQL Task utilizing a T-SQL MERGE statement.

How it Works: The MERGE statement performs the "Upsert" (Update existing records + Insert new records) operation directly inside the SQL Server Database Engine, rather than pulling data into the SSIS memory buffers.

Conclusion

In conclusion, this project has successfully designed and implemented a robust, enterprise-grade ETL framework tailored for the retail industry. By leveraging SQL Server Integration Services (SSIS), we transformed fragmented, raw operational data into a highly structured and optimized Data Warehouse, directly addressing the core business need for reliable and scalable analytical reporting.

The implementation strictly adhered to industry best practices by establishing a multi-tiered architecture encompassing Staging, Data Validation, Business Transformation, and the final Data Warehouse layers. A significant technical achievement of this project was the seamless integration of both Full and Incremental load strategies. Through rigorous performance tuning—such as optimizing Lookup caches, tuning pipeline buffer sizes, and implementing T-SQL MERGE statements for massive fact tables—we successfully overcame critical memory bottlenecks. This ensured the data pipeline is not only highly efficient but also resilient against large-scale data processing challenges.

Furthermore, the incorporation of automated error handling, data quality checks, and centralized logging guarantees data integrity and provides complete audit traceability. The resulting system establishes a reliable "Single Source of Truth" that empowers stakeholders to make data-driven decisions.

Looking ahead, this foundational architecture is fully prepared for future scalability. Potential enhancements include migrating the pipeline to cloud-native services like Azure Data Factory, integrating Change Data Capture (CDC) for near real-time data streaming, and deploying automated data quality dashboards. Ultimately, this project demonstrates a strong command of modern data engineering principles and delivers a production-ready solution that adds immediate strategic value to the business.

Bonus Tasks

Create a Data Quality Dashboard showing: Total Processed Records, Total Valid Records, Total Rejected Records, Validation Success Rate, Execution Duration